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Technical guide
Machine Vision AI

Webfleet Video 2.0

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Technical guide

Webfleet Video 2.0 offers a comprehensive suite of MVAI functionalities in both the CAM Lite and the CAM Pro dash cameras.

This guide helps getting a better understanding of how these features work, and guides through troubleshooting and testing.

Machine Vision Artificial Intelligence

ADAS
Advance Driving Assistance System.
(ADAS) are a set of technologies designed to enhance vehicle safety and improve the driving experience by using sensors, cameras, and artificial intelligence. These systems help prevent accidents, reduce human errors, and assist drivers in various ways.
ADAS improves road safety by reducing the likelihood of accidents and assisting drivers in challenging situations. As technology evolves, these systems are becoming more advanced, paving the way for fully autonomous driving in the future.
ADAS features included in Webfleet Video 2.0: Lane crossing, Tailgating, Time to collision (TTC), Stop and go, Rolling stop (stop sign violations).

DMS
Driver Monitoring System.
(DMS) is an advanced safety technology designed to assess a driver's alertness, attention, and behavior in real-time. Using cameras, infrared sensors, and artificial intelligence, DMS detects signs of fatigue, distraction, or impaired driving to enhance road safety and prevent accidents.
DMS plays a crucial role in reducing accidents caused by driver inattention and fatigue. With increasing adoption in modern vehicles, it is a key step toward safer driving and the future of autonomous mobility.
DMS features included in Webfleet Video 2.0: Distraction, Fatigue, Phone usage, Seatbelt detection, Lens covered.

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Why need to do "DMS Auto Calibration" processing?

Objective

The Driver Monitoring System (DMS) requires auto calibration due to:

1. **Camera Placement Variability:**
   - Each vehicle may have a different camera installation position.
2. **Driver Variability:**
   - Drivers have unique body sizes and driving postures.
3. **Algorithmic Adjustment:**
   - DMS uses algorithms to automatically calculate appropriate trigger parameters, ensuring accurate event detection in subsequent operations.

System Setup

1. **Configuring Driving Orientation**
   - Before using the system, users must correctly set the vehicle's **left-hand drive** or **right-hand drive** configuration through the **installation app**.
2. **Activation Conditions**
   - **Vehicle Speed**: Auto calibration begins only when the vehicle speed exceeds **20 km/h**.
   - **Stable Posture**: Calibration requires the driver to maintain a **steady and consistent seating posture**.
1. **Calibration Procedure**
   - The system calculates installation parameters algorithmically based on the driver's stable posture.
   - **Calibration Time**: Usually completed within 3 minutes, but it may be affected by:
     - Traffic Flow: Smooth traffic is necessary; calibration may be delayed if there is heavy traffic, frequent stops, or red lights.
     - Driver Stability: The driver must maintain a stable posture without excessive body movement.

Limitations and Challenges

1. **Traffic and Driver Behavior**:
   - Calibration may be delayed or fail under these conditions:
     - **Stop-and-go traffic** flow
     - The driver engages in frequent movements, such as:

        - Adjusting the central console
        - Searching for items
        - Turning to talk to passengers
2. **Facial Obstructions**:
    - If the driver wears a **mask** during calibration, the system cannot detect facial features, preventing the calibration from completing successfully.
3. **Weak GPS Signal**:
    - DMS functionality is disabled if the GPS signal is weak or unavailable, such as:
        - **Driving in tunnels**
        - **Indoor parking areas**

Notes

- **Maintain Stability**: Drivers should remain in a steady posture throughout the calibration process.
- **Avoid Movements**: Minimize distractions and unnecessary movements, such as adjusting controls or engaging in conversations.
- **Remove Obstructions**: Ensure the face is fully visible by avoiding masks or other items covering the face during calibration.
- **Choose Optimal Conditions**: Perform calibration in smooth traffic conditions with minimal interruptions to expedite the process.

By adhering to these guidelines, users can ensure accurate calibration of the DMS, enabling optimal performance and reliable event detection tailored to each vehicle and driver's unique setup.

How to Test the "Unfasten Seatbelt" Event?

Objective

To ensure the seatbelt detection model accurately triggers an event when it detects the driver is not wearing a seatbelt.

Testing Steps

1. **Pre-test Conditions**
   - Ensure the vehicle exceeds the **configured speed threshold** (≥ 20 km/h).
   - Confirm the **Driver Monitoring System (DMS) calibration** is completed:
     - **Calibration** typically requires the driver to maintain a stable posture for approximately **3 minutes**.
     - **Vehicle** speed must remain above **20 km/h** during calibration.
2. **Event Trigger Setup**
   - Unbuckle the seatbelt to simulate the "unfasten seatbelt" event.
   - Verify that the **in-cabin camera view** clearly shows the lap band of the seatbelt.
   - Adjust the camera's pitch angle to cover the driver's **head-to-waist area** for optimal detection accuracy.
   - Ensure the driver remains **unbuckled for more than 12 seconds** (medium level) to trigger the event.
3. **Testing Environment**
   - Ensure the seatbelt is not obscured by:
     - **Clothing**
     - **Hands**
     - **Other objects**
   - Avoid testing other functions simultaneously, such as:
     - **Distraction detection**
     - **Fatigue monitoring**
     - **Phone usage detection**
4. **Post-Trigger Interval**
   - After the event is triggered, there is a **30-second debounce interval** during which no new events will be logged.
   - Ensure that repeated testing avoids this interval to properly validate the feature.

Challenges and Considerations

1. **Environmental Factors**:
    - Minimize issues that may reduce detection accuracy, such as:
        - **Image exposure problems**
        - **Blurry images**
        - **Insufficient image brightness**
2. **Clothing Complexity**:
    - Avoid wearing clothing with **complex patterns or excessive details**, as this can challenge the AI model's ability to distinguish the seatbelt, potentially reducing recognition accuracy.
3. **Object Obstruction**
    - Objects covering the seatbelt, such as **hands, clothing, or other items**, may cause **false alerts**.
        - **When** the seatbelt is obscured and does not appear in the image, the AI model may incorrectly assume the seatbelt is not fastened.
4. **Camera Installation Angle**
    - If the camera's **pitch angle** does not capture the complete seatbelt (e.g., only capturing the driver's chest and above), this presents a significant challenge for the AI model. The detection accuracy may become unstable or unreliable.

Notes

- **Camera Positioning**: Proper adjustment of the camera pitch angle enhances the detection model's performance by ensuring the seatbelt is clearly visible.
- **Calibration Period**: Ensure the calibration process is completed before testing. The seatbelt detection feature will only activate after this step.
- **Lighting Conditions**: Perform testing in environments with adequate lighting to avoid recognition errors.
- **Clothing Recommendations**: Simple, solid-colored clothing is recommended during testing to improve the model's ability to detect the seatbelt accurately.

By following these steps, the "Unfasten Seatbelt" event can be thoroughly tested to help ensure that the functionality meets intended specifications. However, please note that under challenging real-world conditions—such as complex clothing patterns, poor lighting, or other environmental factors—the model may not always achieve a high level of accuracy.

Phone Usage Detection Testing Manual

How to Test the "Phone Usage" Event?

Objective

To ensure the phone usage detection model accurately triggers an event when the driver is using a phone by holding it to their ear or interacting with the screen.

Testing Steps

1. **Pre-test Conditions**
   - Ensure the vehicle exceeds the **configured speed threshold** (≥ 20 km/h).
   - Confirm the **Driver Monitoring System (DMS) calibration** is completed:
     - Calibration typically requires the driver to maintain a stable posture for approximately **3 minutes**.
     - Vehicle speed must remain above **20 km/h** during calibration.
2. **Event Trigger Setup**
   - Perform one of the following actions:
     - **Hold the phone close to your ear** to simulate a call.
     - **Hold the phone and slide your finger on the screen** to simulate interaction.
   - Ensure the phone is held in your hand during the test:
     - Avoid mimicking phone usage with other objects, as this could result in **false alarms**.
   - Continue using the phone for **more than 7 seconds** (medium level) to trigger the event.
1. **Camera Visibility**
   - Ensure the hands and phone are **clearly visible** within the camera's field of view.
   - The event will not be detected if:
     - The phone or hands are obscured.
     - Visibility is compromised by objects or poor lighting.
2. **Post-Trigger Interval**
   - After the event is triggered, there is a **30-second debounce interval** during which no new events will be logged.
   - Ensure that repeated testing avoids this interval to properly validate the feature.

Challenges and Considerations

1. **Small Object Detection**:
    - Since the phone is a small object in the camera's field of view, the AI model may face challenges in accurately identifying it.
    - Actions such as **touching the face** or **holding other objects (e.g., boxes, electric shavers, drink cartons, or walkie-talkies)** in a manner that mimics phone usage could lead to **false detections**.
    - These behaviors may affect the recognition accuracy of the AI model.
2. **Environmental Factors**:
    - Ensure the test environment provides sufficient lighting and unobstructed views to minimize detection errors.

Notes

- **Calibration Period**: Ensure the calibration process is completed before testing. The phone usage detection feature will only activate after this step.
- **Testing Duration**: Ensure phone usage lasts for at least 10 seconds to reliably trigger the event.
- **Camera Positioning**: Adjust the camera’s pitch angle to include the driver’s hand movements within the detection zone.
- **Behavioral Variations**: Avoid actions that mimic phone usage with unrelated objects (e.g., **electric shavers, drink cartons, or walkie-talkies**) to maintain test integrity.

By following these guidelines, you can thoroughly test LDWS events and ensure reliable, accurate performance under ideal road conditions.

How to Test the "Distraction" Event?

Objective

To ensure the distraction detection model accurately triggers an event when the driver exhibits head-turn behaviors indicative of distraction.

Testing Steps

1. **Pre-test Conditions**
   - Ensure the vehicle exceeds the **configured speed threshold** (≥ 20 km/h).
   - Confirm the **Driver Monitoring System (DMS) calibration** is completed:
     - Calibration typically requires the driver to maintain a stable posture for approximately **3 minutes**.
     - Vehicle speed must remain above **20 km/h** during calibration.
2. **Event Trigger Setup**
   - Simulate distraction by turning the head **left** or **right** for a duration exceeding the configured detection time **more than 7 seconds** (medium level).
   - Avoid turning the head briefly as it may not trigger the event.
3. **Post-Trigger Interval**
   - After the event is triggered, there is a **30-second debounce interval** during which no new events will be logged.
   - Ensure that repeated testing avoids this interval to properly validate the feature.

Challenges and Considerations

1. **Face Obstruction**:
   - If the driver is wearing **sunglasses** or **masks**, or if the face is partially obscured by **hands** or other objects, the model may face challenges, potentially leading to:
     - **Missed detections**
     - **False alerts**
2. **Camera Installation**:
   - The position of the camera significantly impacts the detection accuracy:
     - **Optimal Position**: The camera should be installed as close as possible to the driver's **frontal area** to ensure clear visibility of head movements.

        - **Potential Issues**: If the driver's head turns in the **opposite direction** of the camera, the AI model may not correctly detect the angle, reducing accuracy.
3. **Lighting Challenges**:
    - **Bright sunlight** may cause overexposure of facial features, making them unrecognizable to the AI model and potentially disabling the functionality.
4. **Driver Posture Adjustments**:
    - The system relies on **auto-calibration** to establish the driver's normal posture. Changes in posture during driving may cause:
        - **False distraction alerts**
        - Temporary inaccuracies until the system recalibrates (typically within a few minutes).
    - During testing, avoid:
        - **Sudden posture changes**
        - **Frequent head movements**, such as turning to talk to passengers, to maintain consistent test results.
5. **Driving Scenarios**:
    - Lane changes may require the driver to **turn their head to check side mirrors**, potentially causing **false alerts**. This limitation is due to the current AI model's challenge in differentiating intentional mirror-checking from distracted behavior.

Notes

- **Calibration**: Ensure the system has completed auto-calibration before starting the test.
- **Test Stability**: Maintain a consistent driving posture throughout the test to avoid unnecessary recalibration.
- **Event Trigger Limitations**: Be mindful of scenarios like lane-changing or face obstructions that could impact the detection system's accuracy.

By following these guidelines and taking relevant situational factors, the "Head-Turn Distraction" detection feature can be effectively tested, helping ensure the system's accuracy and reliability.

How to Test the "Fatigue Behavior" Event?

Objective

To ensure the fatigue detection model accurately triggers an event when the driver exhibits fatigue-related behaviors, such as head-drooping, head-tilting backward, or prolonged eye closure.

Testing Steps

1. **Pre-test Conditions**
   - Ensure the vehicle exceeds the **configured speed threshold** (≥ 20 km/h).
   - Confirm the **Driver Monitoring System (DMS) calibration** is completed:
     - **Calibration** typically requires the driver to maintain a stable posture for approximately **3 minutes**.
     - **Vehicle** speed must remain above **20 km/h** during calibration.
2. **Event Trigger Setup**
   - Simulate fatigue by performing one of the following actions for **more than 15 nods within one minute or 10 seconds with eyes closed** (medium level):
     - **Combination of Head Movements and Eye Closure**: **Drooping the head** (looking downward) **or Tilting the head backward** (napping posture), **combined with closing the eyes**. One nod is considered after 2 seconds of head looking downward tilting head backward.
     - **Eye Closure Alone**: Even without head movements, the model will trigger an event if the **eyes are closed**, provided the **environmental lighting is bright**.
   - Lighting Conditions:
     - Brightness is automatically determined by the device based on the camera exposure time.
     - Bright Light: Both **head movements and eye closure** are required to trigger the event if head-drooping or tilting occurs.
     - Low Light: Independent eye closure detection is disabled, and the model will rely on **head movements plus eyes closure** (drooping or tilting backward) for the trigger condition.
3. **Post-Trigger Interval**
   - After the event is triggered, there is a **30-second debounce interval** during which no new events will be logged.
   - Ensure repeated testing avoids this interval to validate the feature properly.

Challenges and Considerations

1. **Face Obstruction**:
    - The model may encounter challenges if the driver is wearing **sunglasses**, **masks**, or if the face is partially obscured by **hands** or **other objects**, potentially leading to:
        - **Missed detections**
        - **False alerts**
2. **Camera Installation**:
    - The camera's position significantly affects detection accuracy:
        - **Optimal Position**: Ensure the camera can clearly capture the driver's face, especially when the head is tilted downward or backward.
        - **Potential Issues**: If the **head-tilt angle is too extreme**, facial features may move out of the camera's field of view, reducing accuracy.
3. **Eye Condition Assessment**:
    - Bright Light: Eye state detection is reliable in bright conditions when facial features are clearly visible.
    - Low Light: Due to poor pixel quality of the eyes in low-light environments, independent **eye state assessment is not performed at night**. The system relies on combinations of other fatigue indicators such as head-drooping or tilting behaviors instead.
4. **Lighting Challenges**:
    - **Bright sunlight** may cause overexposure of facial features, making them unrecognizable to the AI model and potentially disabling the functionality.
5. **Driver Posture Adjustments**:
    - The system relies on **auto-calibration** to determine the driver's normal posture. Sudden posture changes during driving may cause:
        - **False fatigue alerts**
        - Temporary inaccuracies until recalibration (typically within a few minutes).
    - During testing, avoid:
        - **Sudden posture changes**
        - **Frequent head movements**, such as turning to talk to passengers, to maintain consistent test results.
6. **Behavioral Variability**:
    - Actions such as **looking down to pick up objects** or other distracted behaviors may mimic fatigue, leading to **false alerts**. This is a limitation of the current AI model.

Notes

- **Calibration Period**: Ensure the system completes auto-calibration before starting the test.
- **Testing Stability**: Maintain a consistent driving posture to avoid unnecessary recalibration and false alerts.
- **Lighting Conditions**:
    - **Bright Light**: Eye closure detection is prioritized and works in conjunction with head movements for triggers.

- **Environmental Conditions**: Perform testing in well-lit environments and avoid situations with excessive sunlight or poor lighting that can impact recognition accuracy.

By following these steps and taking into account the limitations that arise during nighttime testing, the "Fatigue Behavior" detection feature can be thoroughly evaluated, thereby increasing confidence in its reliability and accuracy under a wide range of conditions.

Function Temporarily Disabled (DMS Off) Testing Manual

Objective

To verify that the Driver Monitoring System (DMS) correctly triggers a "Function Temporarily Disabled" event when camera abnormalities occur and records the event for diagnostic purposes.

Testing Steps

1. Pre-test Conditions
   - **Vehicle Speed**: Ensure the vehicle exceeds the configured speed threshold (≥ 20 km/h).
2. Event Trigger Setup
   - **Trigger Conditions:**
     - **Camera Obstruction**: Block the camera lens partially or fully.
     - **Driver Out-of-Range**: Move the driver out of the camera's field of view.
3. Trigger Timing
   - The system will trigger the **"Function Temporarily Disabled" event** if the driver cannot be detected continuously for **1 minute**.
4. Trigger Frequency
   - To conserve **network bandwidth**, the system applies an **exponential backoff** for event notifications:
     - **Time** intervals between event notifications increase by the power of 2 (in minutes):**1 minute, 2 minutes, 4 minutes, 8 minutes, 16 minutes, 32 minutes**, and so on.

Notes

- **Calibration**: Ensure DMS calibration is completed and stable before starting the test.
- **Testing Environment**: Use realistic scenarios such as temporary camera obstructions.
- **Event Monitoring**: Verify that events are recorded with timestamps and at the correct intervals according to the exponential strategy.

By following these steps, the "Function Temporarily Disabled" event can be thoroughly tested to ensure accurate detection of camera abnormalities and efficient event logging while conserving network bandwidth.

Why need to do "ADAS Auto Calibration" processing?

Objective

Advanced Driver Assistance Systems (ADAS) requires auto calibration due to:

Each vehicle may have a unique camera installation height and angle, based on its design and mounting location. As a result, an auto-calibration process is required to calculate the correct camera parameters. This ensures that ADAS can accurately gauge the positions (distances) of surrounding objects on the road, providing reliable and timely alert predictions.

System Setup

1. **Camera Installation Check**
   - **Installation App**: Use the app to ensure the camera's lens angle is positioned correctly.
   - **Vanishing Point (Horizon)**: Ideally, this should appear around halfway down the vertical axis of the image.
   - **Green Rectangular Guide**: A visual aid in the app's preview screen helps align the camera properly.
2. **Activation Conditions**
   - **Vehicle Speed**: Auto calibration begins only when the vehicle speed exceeds **30 km/h**.
   - **Stable GPS signal**: ADAS calibration may be disabled if the GPS signal is weak (e.g., inside tunnels, near tall buildings, or in mountainous areas).
1. **Calibration Procedure**
   - **Road Conditions**: Perform calibration when road lanes are clearly visible on a smoothly flowing road. **Avoid poor weather** conditions (heavy rain, snow, or glare) that may obscure lane markings.
   - **Calibration Time**: Auto calibration typically completes within 10 minutes, depending on road conditions.
   - **Smooth Traffic Flow**: Conduct **calibration on highways** or roads with **minimal congestion** and fewer red lights for best results. This shortens calibration time and improves accuracy.

Limitations and Challenges

1. **Poor Road Quality**:
   - Adverse Weather: Snow or rain can create puddles or glare on the road.

    - Faded/Unclear Lane Markings: Worn-out lane stripes can reduce detection accuracy.
    - Shadows: Reduced contrast from shadows may obscure lane lines.
    - These factors can extend calibration time or lead to calibration failure.
2. **Windshield Reflections**:
    - Dashboard Items: Reflections or glare caused by objects on the dashboard can interfere with lane detection.
    - Calibration Impact: Such interference can affect calibration accuracy and prolong the process.

Notes

- **Optimal Conditions**: Maintain a speed above 30 km/h with a stable GPS signal on clear roads and in good weather to ensure successful calibration.
- **Camera Alignment**: Use the Installation App to center the horizon around the image midpoint, following the green guide for proper lens angle.
- **Time and Environment**: Calibration typically finishes in about 10 minutes, though adverse traffic or weather may cause delays.
- **Smooth Traffic Flow**: Calibrate on highways or roads with minimal congestion and fewer red lights to shorten calibration time and improve accuracy.
- **Limitations**: Reflections, worn lane markings, and extreme weather can reduce calibration accuracy or lead to failures.

By understanding these guidelines and challenges, users can set up ADAS Auto Calibration more effectively, ensuring reliable lane detection and accurate alerting functions.

How to Test the "LDWS" Event (Lane crossing)?

Objective

To monitor the driver's driving behavior and determine if the vehicle is deviating from its lane, triggering a real-time alert.

Testing Steps

1. **Pre-test Conditions**
    - Ensure the vehicle exceeds the **configured speed threshold** (≥ 60 km/h).
    - Confirm the **ADAS calibration** is completed:
        - Auto calibration typically completes **within 10 minutes**, depending on **road conditions**.
        - Vehicle speed must remain above **30 km/h** during calibration.
1. **Event Trigger Setup**
    - Perform one of the following actions:
        - The event will be triggered when the **vehicle leaves the lane** and **stays on the lane markings** for a period of time.
    - **Maintain** this lane-departing position for at least **5 seconds** (medium level).
1. **Post-Trigger Interval**
    - After the event is triggered, there is a **30-second debounce interval** during which no new events will be logged.
    - Ensure that repeated testing avoids this interval to properly validate the feature.

Challenges and Considerations

1. **Poor Road Quality**:
    - Adverse Weather: Snow or rain can create puddles or glare on the road.
    - Faded/Unclear Lane Markings: Worn-out lane stripes can reduce detection accuracy.
    - Shadows: Reduced contrast from shadows may obscure lane lines.
    - These factors may cause lane line images to become **unclear or blocked**, leading to **missed or delayed detections**.
2. **Windshield Reflections**:
    - Dashboard Items: Reflections or glare caused by objects on the dashboard can interfere with lane detection.

- Such **unexpected interference** can result in **false detections** and lower the overall accuracy of the system.

Notes

- **Road Conditions & Weather**: Test on roads with clear lane markings and minimal adverse weather to reduce missed or delayed detections.
- **ADAS Calibration**: The ADAS calibration is completed and the camera lens is unobstructed for optimal detection accuracy.
- **Function Limitations**: Shadows, reflections, and unclear markings can lower detection accuracy.

By following these guidelines and avoiding severe weather or poor road conditions, you can effectively test LDWS events and ensure reliable, accurate performance under suitable real-world driving scenarios.

How to Test the "FCWS" Event (Time to collision)?

Objective

To monitor whether the driver maintains a safe distance from the vehicle in front based on **Time to Collision (TTC)**. When TTC falls below a specified threshold, the system triggers a **real-time alert**.

Testing Steps

1. **Pre-test Conditions**
   - Ensure the vehicle's speed is within **40–160 km/h** for FCWS testing.
   - Confirm the **ADAS calibration** is completed:
     - Auto calibration typically completes **within 10 minutes**, depending on **road conditions**.
     - Vehicle speed must remain above **30 km/h** during calibration.
   - FCWS supports the recognition of cars, trucks, buses, and common construction vehicles or trailers.
1. **Event Trigger Setup**
   - Collision Time Threshold:
     - The system triggers an event when the **Time to Collision (TTC)** is below **650 ms** (medium level).
   - TTC is derived from the **relative GPS speed** and the **distance** between your vehicle and the vehicle ahead.
2. **Post-Trigger Interval**
   - After the event is triggered, there is a **30-second debounce interval** during which no new events will be logged.
   - Ensure that repeated testing avoids this interval to properly validate the feature.

Challenges and Considerations

1. **Detection Distance**:
   a. Vehicles at **long distances** appear smaller on the camera feed; effective detection range is about **20 meters** for FCWS to reliably identify and track the vehicle ahead.
2. **Windshield Reflections**:

a. Dashboard Items: Reflections or glare caused by objects on the dashboard can interfere with lane detection.
b. Unexpected glare may lead to **false detections** or **reduced accuracy** in determining TTC.

Notes

- **Safety Precautions**: Testing tailgating involves driving closer to the vehicle ahead—always be prepared for sudden stops or changes in traffic.
- **Road Selection**: Use clear roads with minimal traffic to allow consistent tracking and reduce the risk of rapid speed changes.
- **Camera Clarity**: Keep the windshield and camera lens free from dirt, smudges, or obstructions.

By following these guidelines, you can systematically test Tailgating events and gather data to refine detection accuracy under various real-world conditions.

How to Test the "Tailgating" Event?

Objective

To monitor whether the driver is maintaining a safe following distance. The system uses **Time to Collision (TTC)** and **prolonged close following** as indicators to trigger a real-time alert if the vehicle remains too close to another vehicle for an extended period.

Testing Steps

1. **Pre-test Conditions**
    - Ensure the vehicle's speed is within **40–160 km/h** for Tailgating testing.
    - Confirm the **ADAS calibration** is completed:
        - Auto calibration typically completes **within 10 minutes**, depending on **road conditions**.
        - Vehicle speed must remain above **30 km/h** during calibration.
    - Tailgating supports the recognition of cars, trucks, buses, and common construction vehicles or trailers.
1. **Event Trigger Setup**
    - Tailgating Time Threshold:
        - The system triggers a Tailgating event when **TTC** is below **650 ms** (medium level), **and** remains below this threshold for **15 seconds** (medium level).
    - TTC is derived from the **relative GPS speed** and the **distance** between your vehicle and the vehicle ahead.
2. **Post-Trigger Interval**
    - After the event is triggered, there is a **30-second debounce interval** during which no new events will be logged.
    - Ensure that repeated testing avoids this interval to properly validate the feature.

Challenges and Considerations

1. **Detection Distance**:
    a. Vehicles at **long distances** appear smaller on the camera feed; effective detection range is about **20 meters** for Tailgating to reliably identify and track the vehicle ahead.
2. **Windshield Reflections**:

a. Dashboard Items: Reflections or glare caused by objects on the dashboard can interfere with lane detection.
b. Unexpected glare may lead to **false detections** or **reduced accuracy** in determining TTC.

Notes

- **Safety Precautions**: Testing tailgating involves driving closer to the vehicle ahead—always be prepared for sudden stops or changes in traffic.
- **Road Selection**: Use clear roads with minimal traffic to allow consistent tracking and reduce the risk of rapid speed changes.
- **Camera Clarity**: Keep the windshield and camera lens free from dirt, smudges, or obstructions.

By following these guidelines, you can systematically test Tailgating events and gather data to refine detection accuracy under various real-world conditions.

Function Temporarily Disabled (ADAS Off) Testing Manual

Objective

To verify that the Advanced Driver Assistance Systems (ADAS) correctly triggers a "Function Temporarily Disabled" event when camera abnormalities occur and records the event for diagnostic purposes.

Testing Steps

1. Pre-test Conditions
   - **Vehicle Speed**: Ensure the vehicle exceeds the configured speed threshold (≥ 30 km/h).
2. Event Trigger Setup
   - **Camera Installation Verification:**
     - Wait for the auto-calibration process to complete.
     - If the **vanishing point (horizon)** is detected **too high** or **too low** on the image's vertical axis, it indicates an improper camera installation angle.
     - In such cases, ADAS functionality will be **temporarily disabled** to avoid generating excessive false alerts.
   - Road Environment Validation:
     - During auto-calibration, the system must detect clear lane markings within 600 seconds.
     - If **adverse conditions** (e.g., weather issues, heavily worn or invisible lane lines) prevent lane detection, calibration will exceed the 600-second threshold.
     - When this timeout is reached, ADAS is **temporarily disabled** to prevent unreliable operation.

Notes

Below are several test methods you can use to trigger and verify the "Function Temporarily Disabled" (ADAS Off) event:

- **Incorrect Camera Angle**: Tilt the camera so the horizon is too high or too low, causing ADAS to disable during auto-calibration.
- **Poor Road/Lane Conditions**: Drive on roads with minimal or obscured markings; ADAS disables if it cannot detect lanes within 600 seconds.

By following these steps, the "Function Temporarily Disabled" event can be thoroughly tested to ensure accurate detection of camera abnormalities and efficient event logging while conserving network bandwidth.

Debug Mode

- **Debug mode** can be enabled through 2line – Webfleet Development Department.
- **Debug mode enabled** will show the AI data and working logic algorithm on the footages and images uploaded to Webfleet. This option is recommended for a greater visibility during troubleshooting and testing.

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